

Introduction to Neural Network

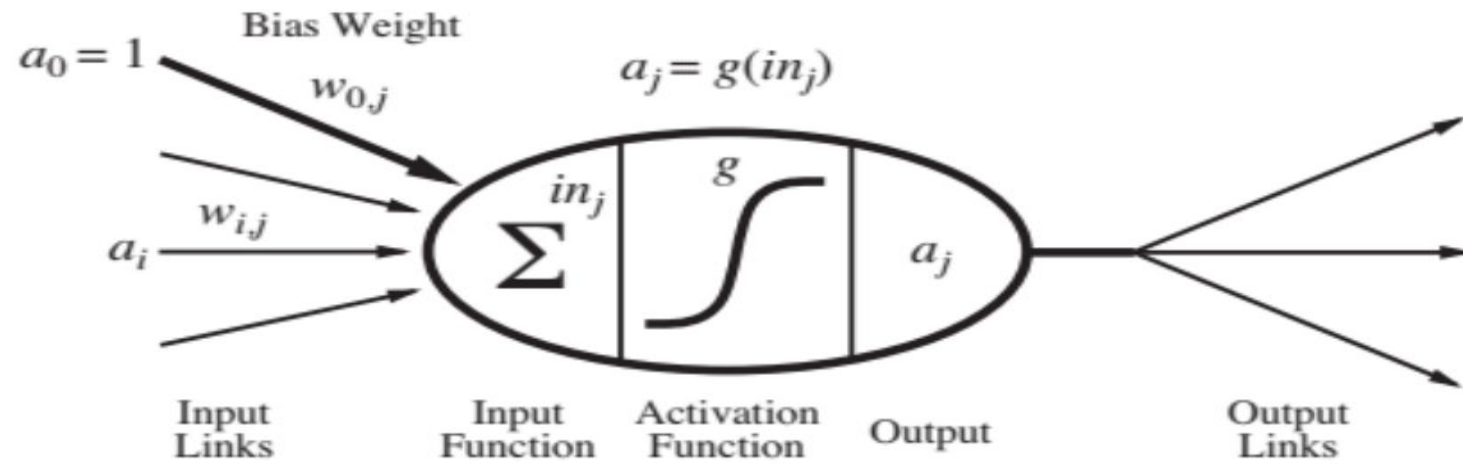
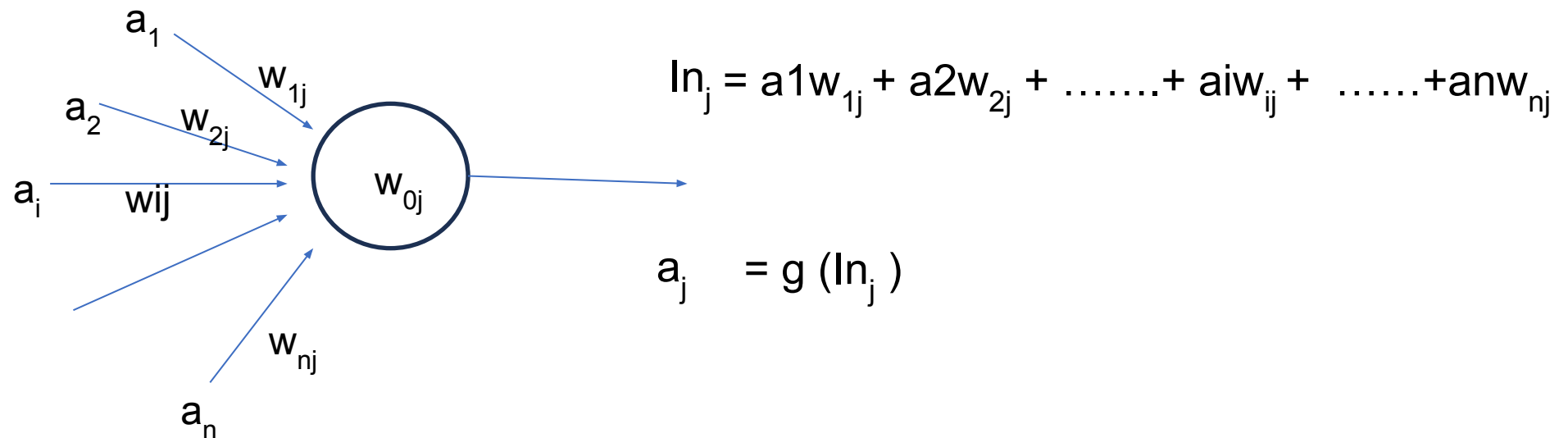


Figure 18.19 A simple mathematical model for a neuron. The unit's output activation is $a_j = g(\sum_{i=0}^n w_{i,j} a_i)$, where a_i is the output activation of unit i and $w_{i,j}$ is the weight on the link from unit i to this unit.



Neural networks are composed of nodes or **units** (see Figure 18.19) connected by directed **links**. A link from unit i to unit j serves to propagate the **activation** a_i from i to j .⁸ Each link also has a numeric **weight** $w_{i,j}$ associated with it, which determines the strength and sign of the connection. Just as in linear regression models, each unit has a dummy input $a_0 = 1$ with an associated weight $w_{0,j}$. Each unit j first computes a weighted sum of its inputs:

$$in_j = \sum_{i=0}^n w_{i,j} a_i .$$

Then it applies an **activation function** g to this sum to derive the output:

$$a_j = g(in_j) = g \left(\sum_{i=0}^n w_{i,j} a_i \right) . \tag{18.9}$$

Feedforward neural network

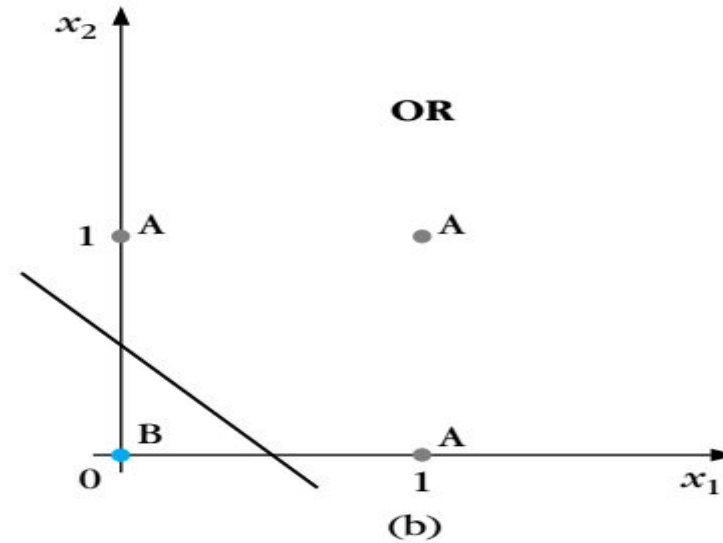
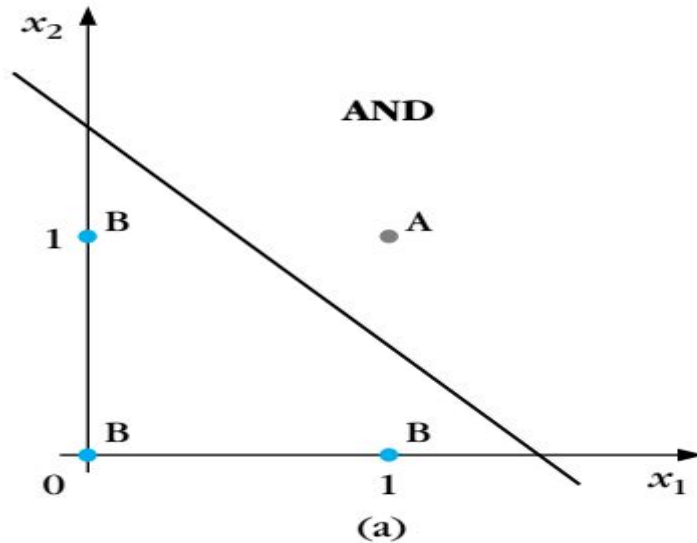
Directed acyclic graph. Only forward connection, no feedback loop. Output depends on current input and weights of the network

Recurrent neural network

Feedback connection is available. Output depends on input and previous state. Suitable for short term memory

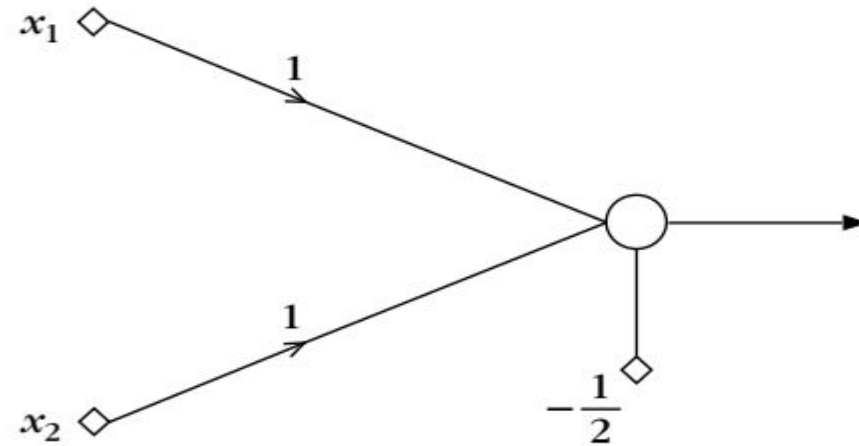
A single neuron acts as a linear classifier and can implement only linearly separable function

Linearly separable function

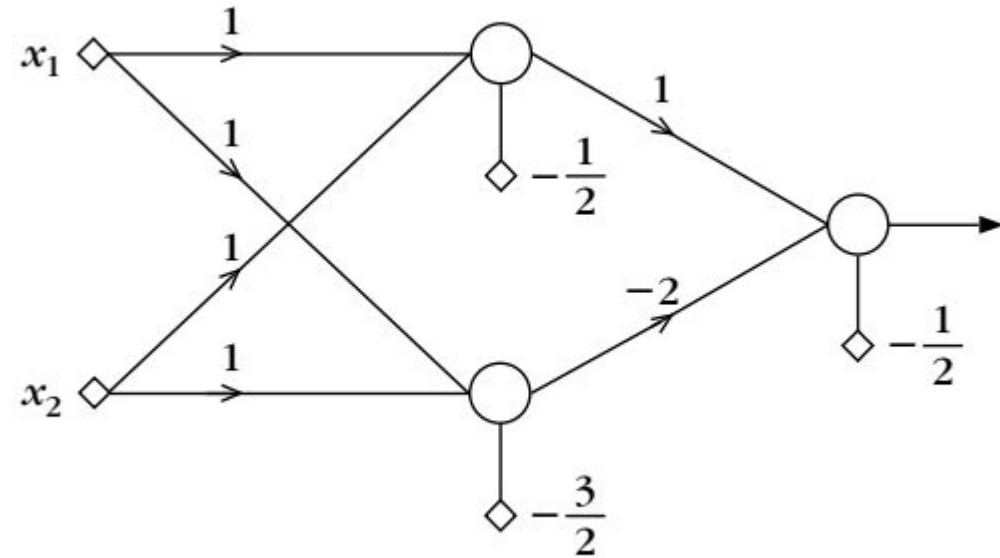
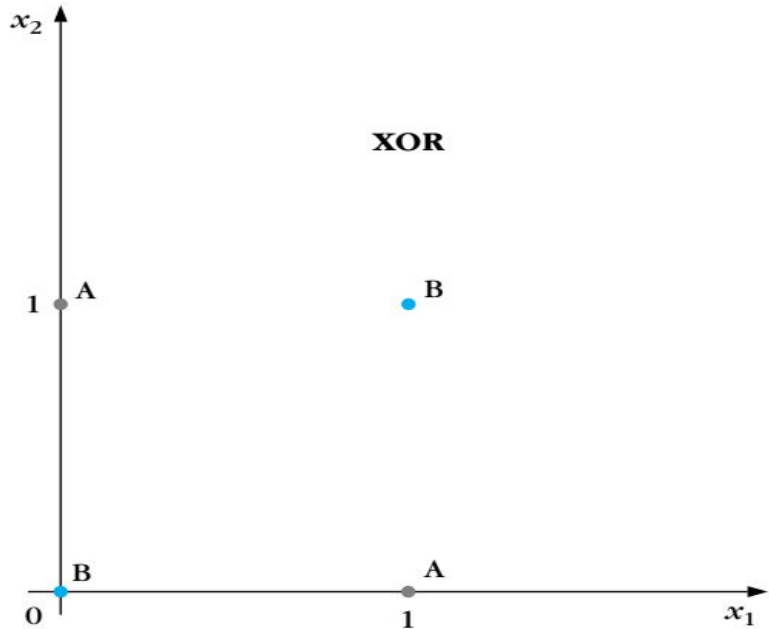


The OR gate implementation →

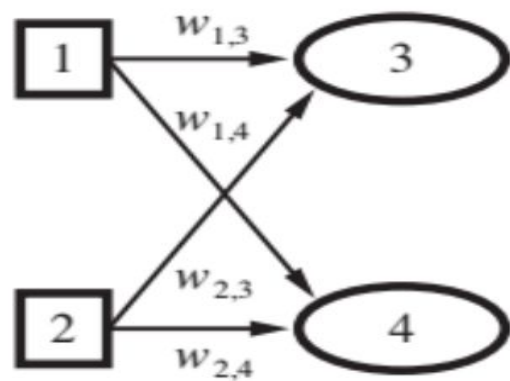
If you change the bias to -1.5
it becomes an AND gate



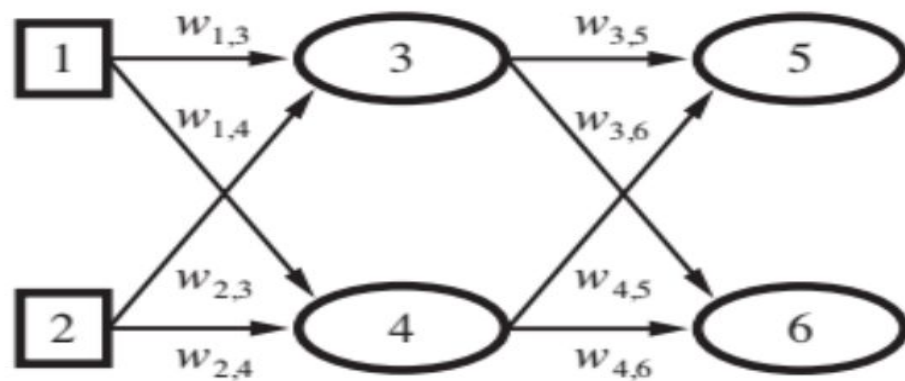
Linearly non separable function



To implement a linearly non separable function, we need multi layer neural network. The above is the implementation of linearly non separable XOR function



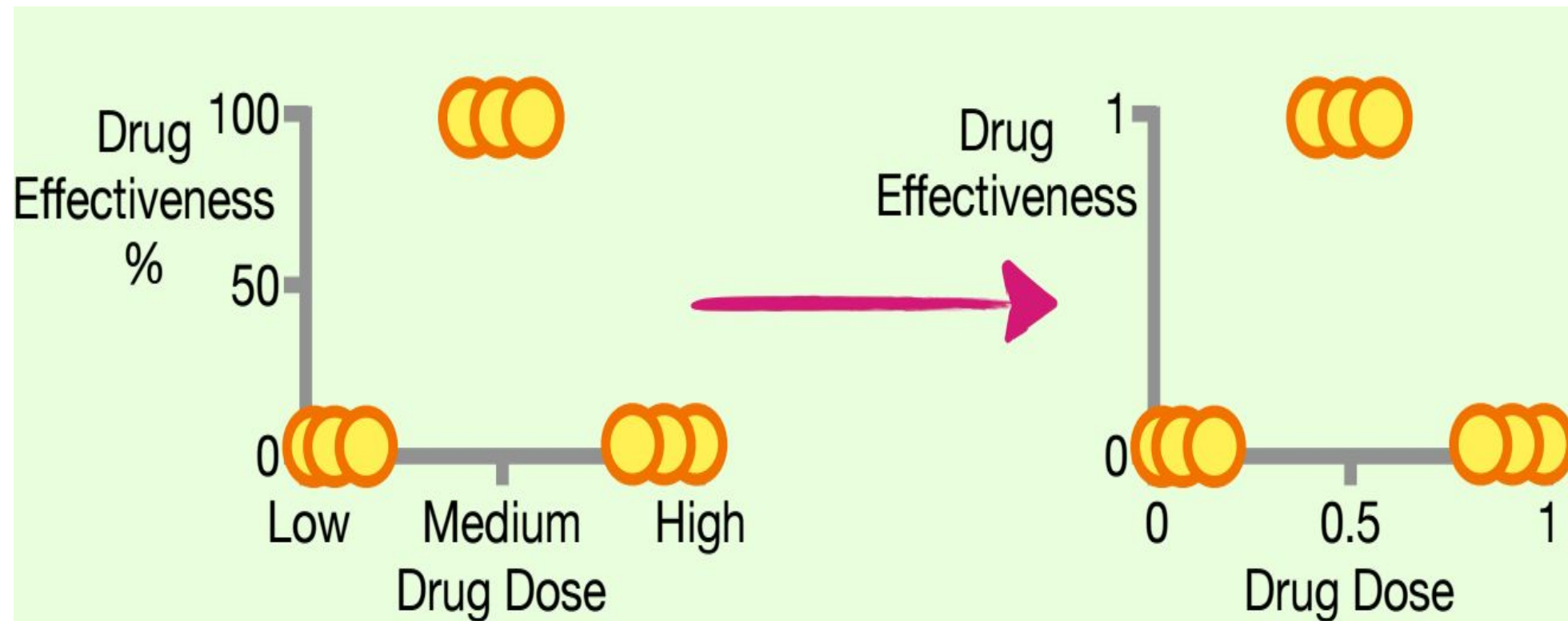
(a)

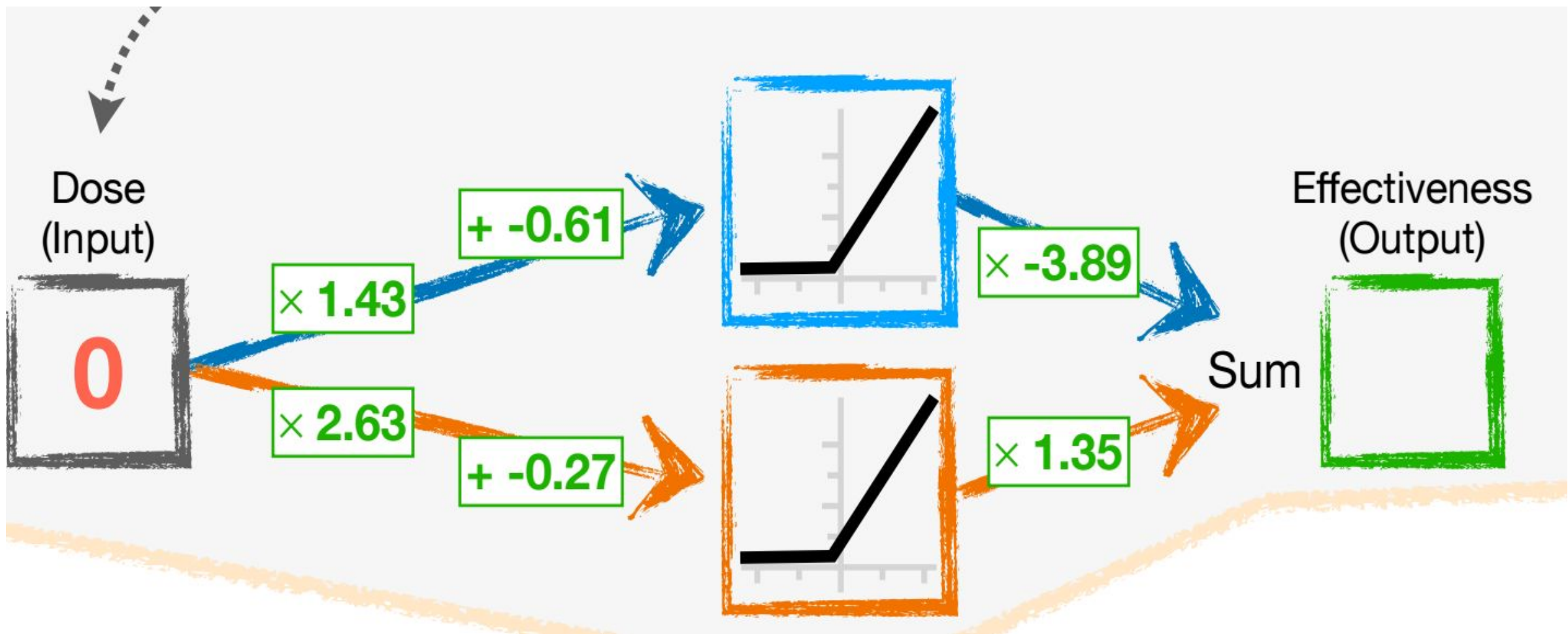


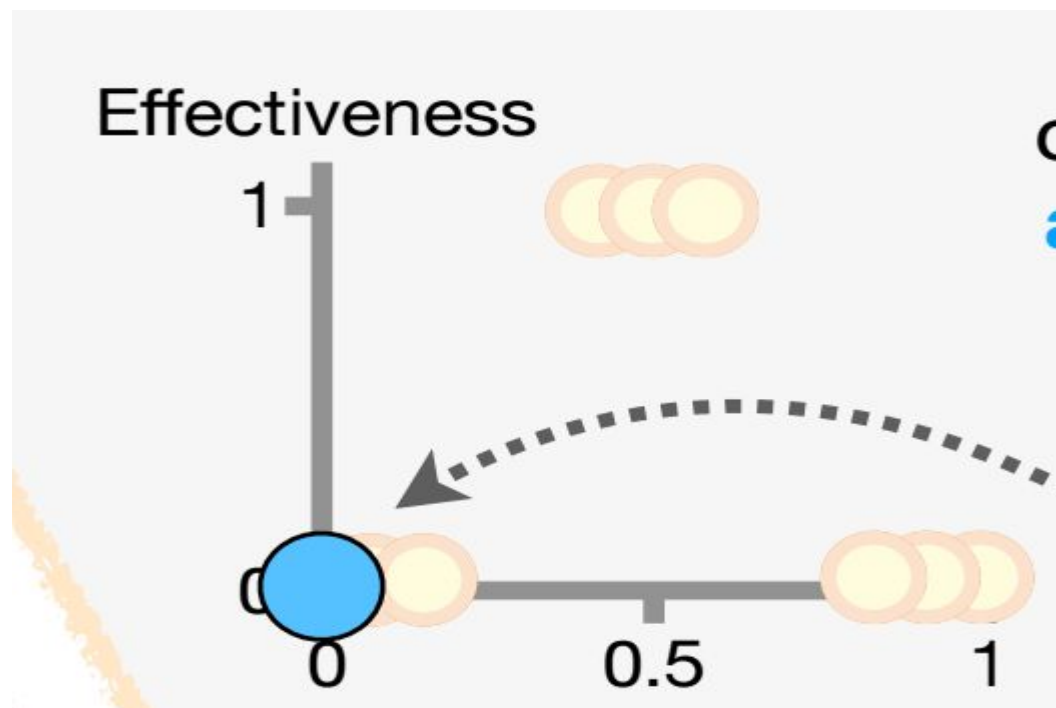
(b)

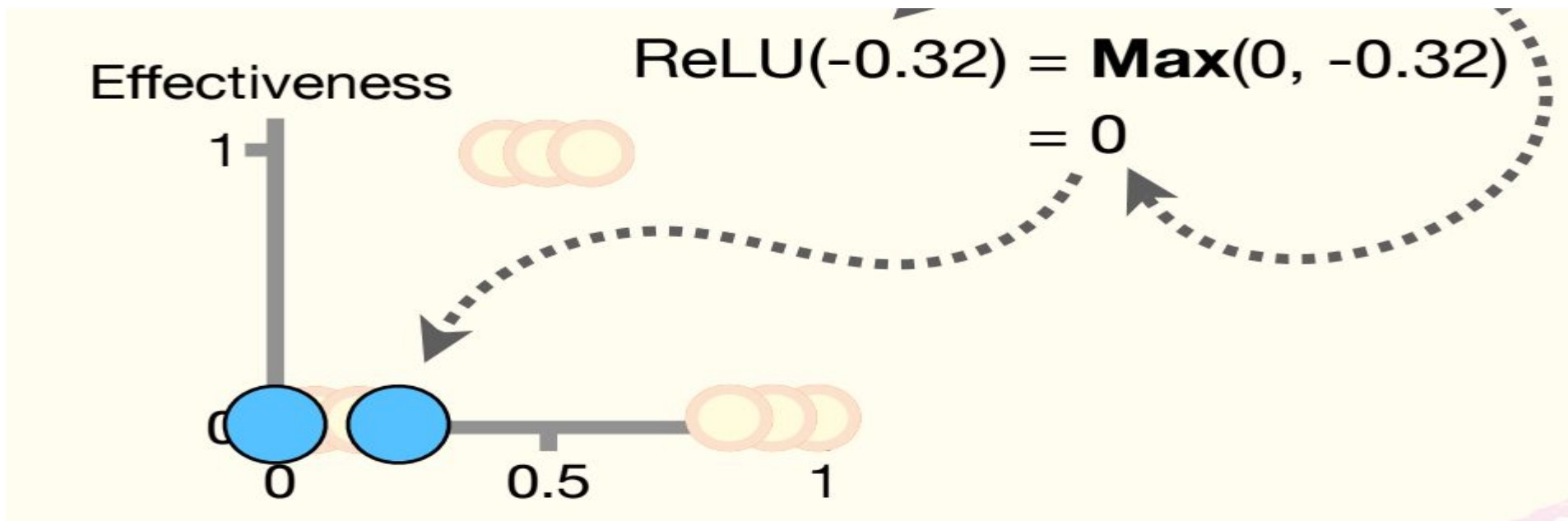
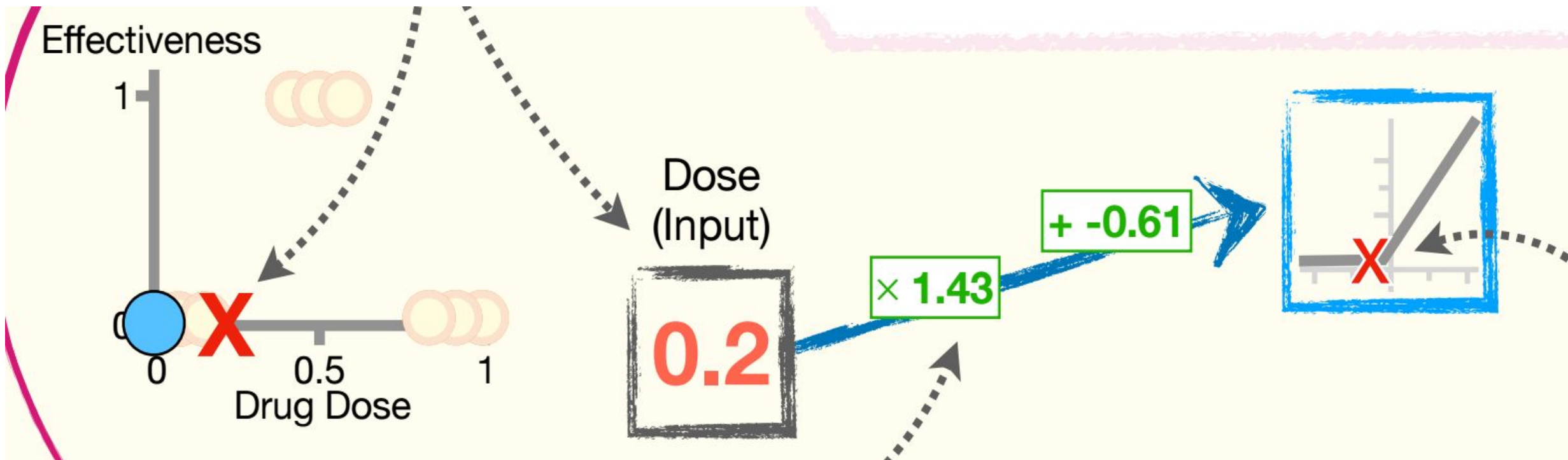
Figure 18.20 (a) A perceptron network with two inputs and two output units. (b) A neural network with two inputs, one hidden layer of two units, and one output unit. Not shown are the dummy inputs and their associated weights.

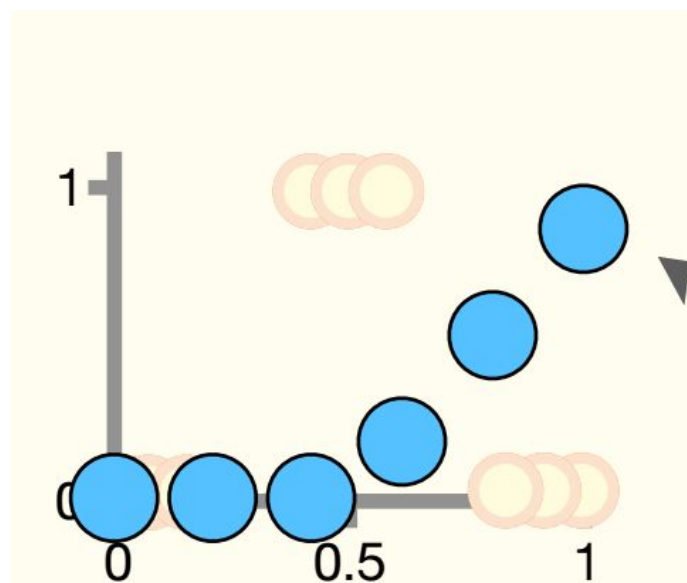
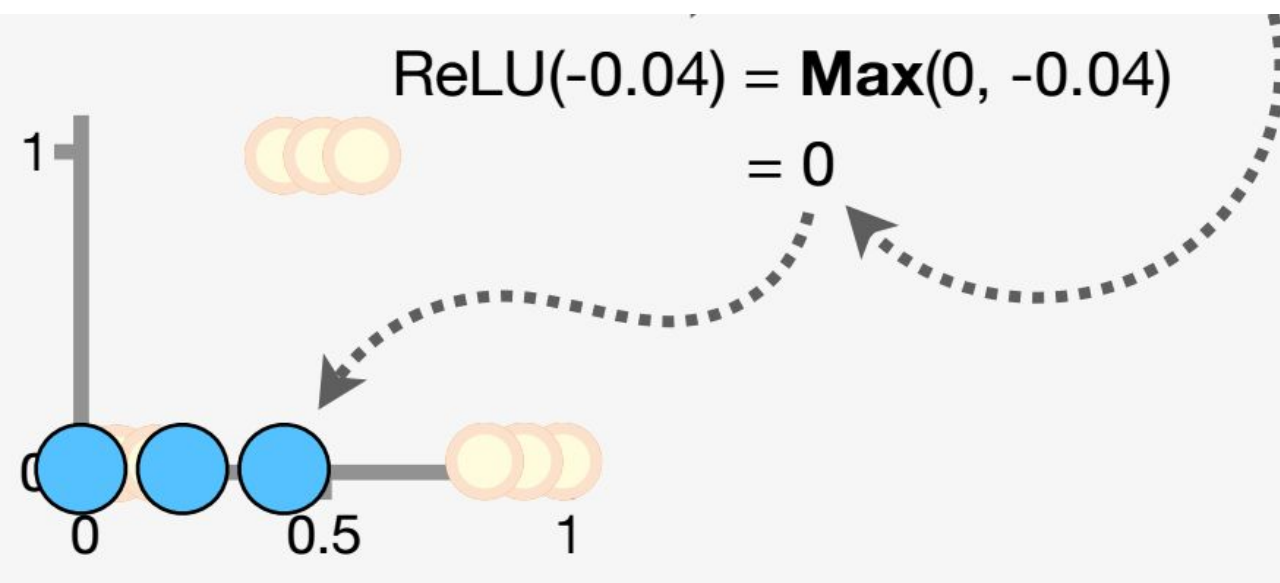
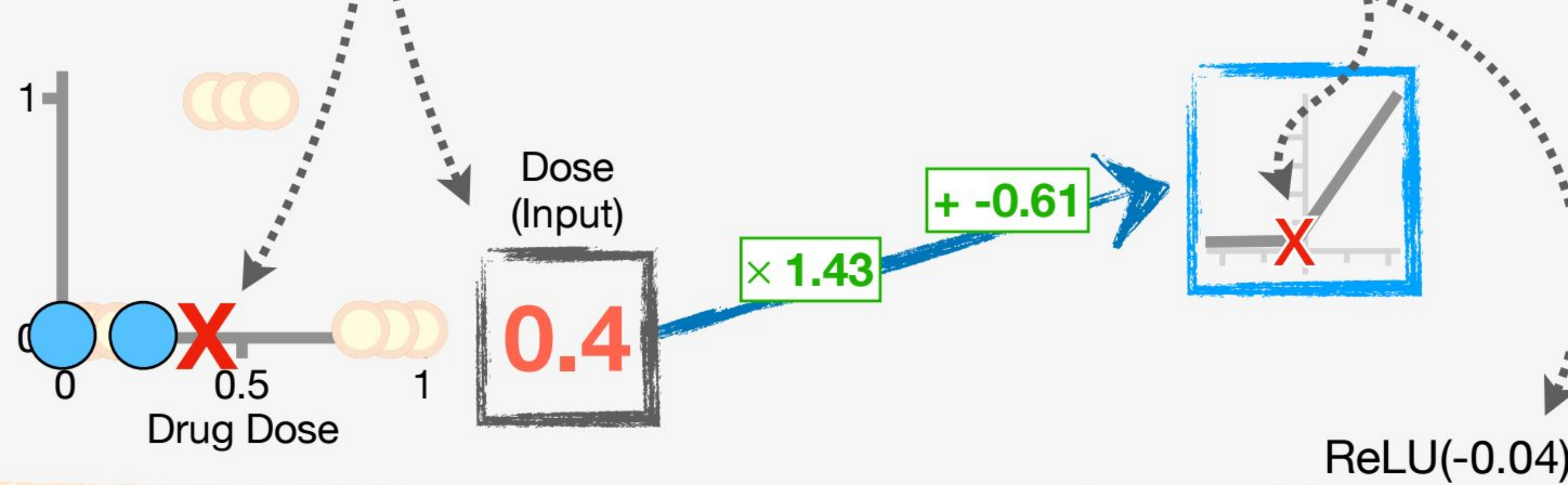
$$\begin{aligned}
 a_5 &= g(w_{0,5} + w_{3,5} a_3 + w_{4,5} a_4) \\
 &= g(w_{0,5} + w_{3,5} g(w_{0,3} + w_{1,3} a_1 + w_{2,3} a_2) + w_{4,5} g(w_{0,4} + w_{1,4} a_1 + w_{2,4} a_2)) \\
 &= g(w_{0,5} + w_{3,5} g(w_{0,3} + w_{1,3} x_1 + w_{2,3} x_2) + w_{4,5} g(w_{0,4} + w_{1,4} x_1 + w_{2,4} x_2)).
 \end{aligned}$$



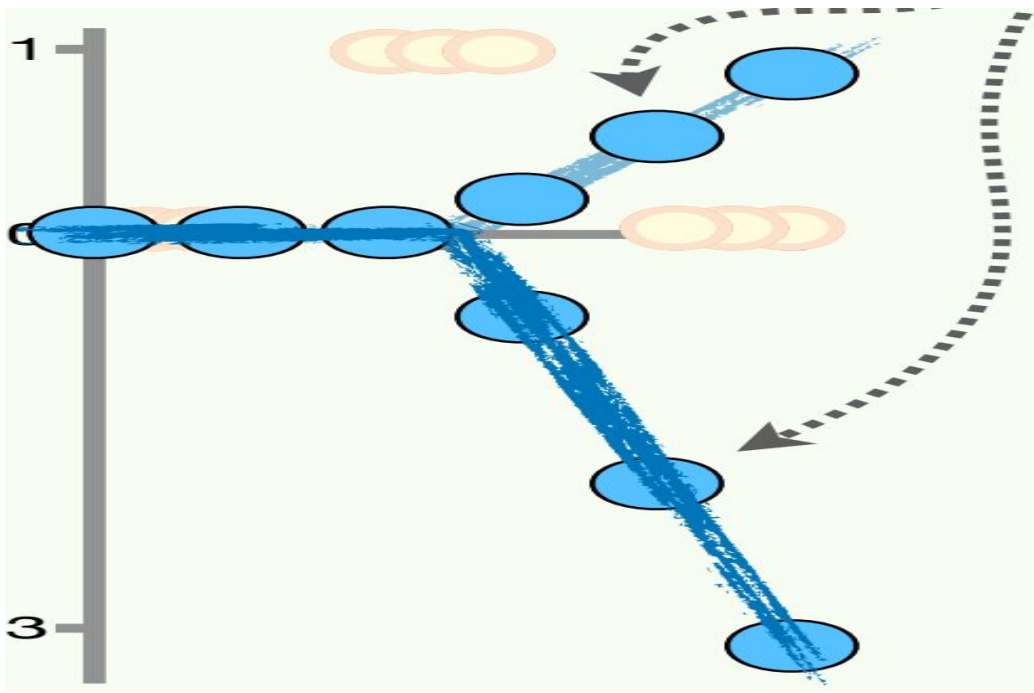
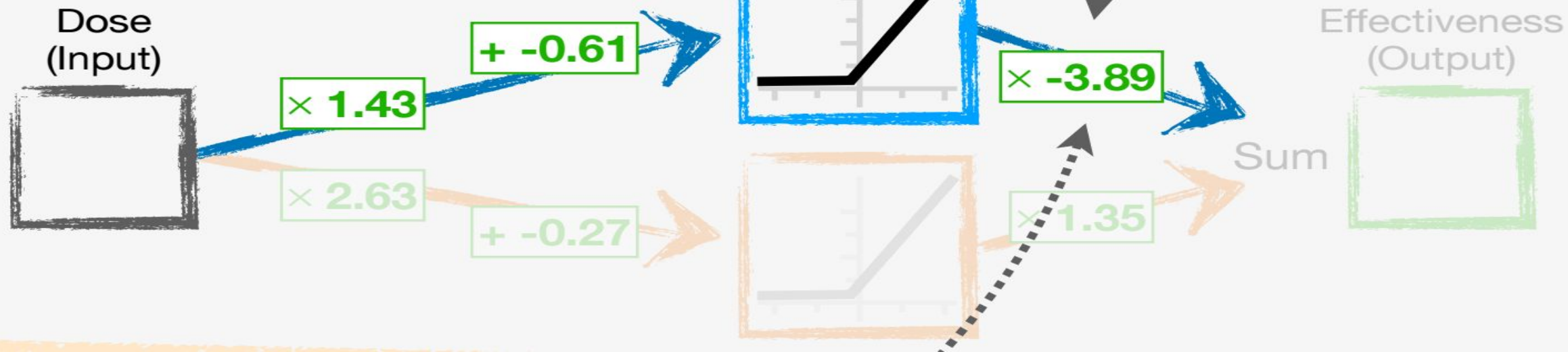


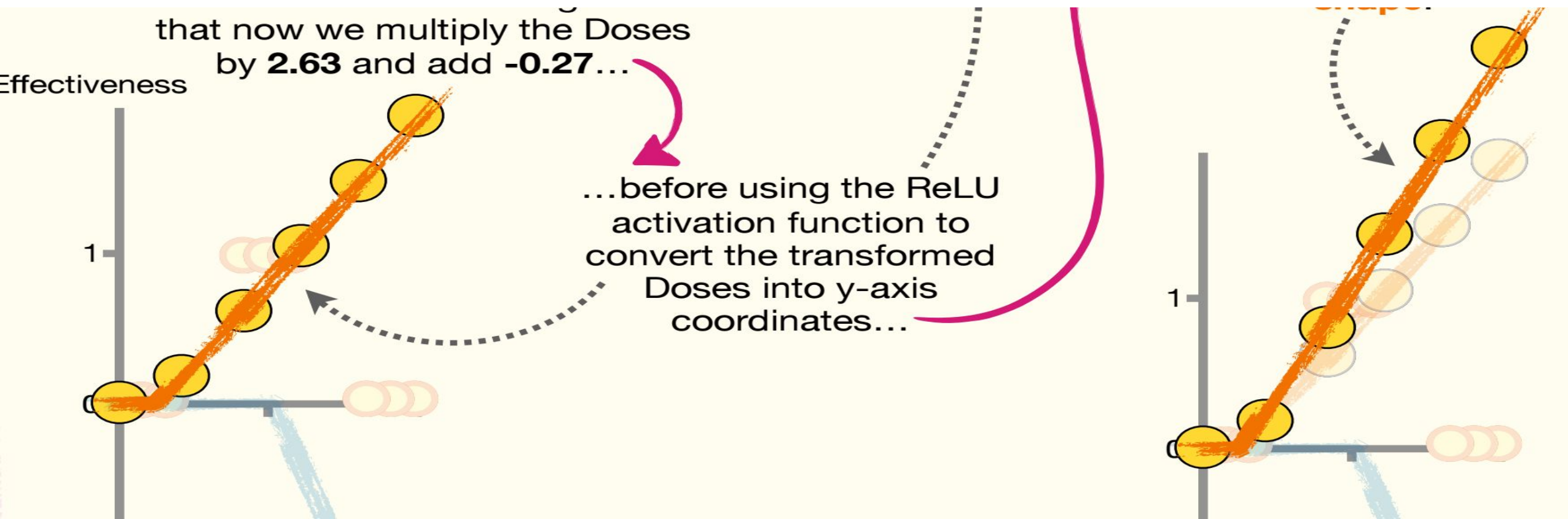


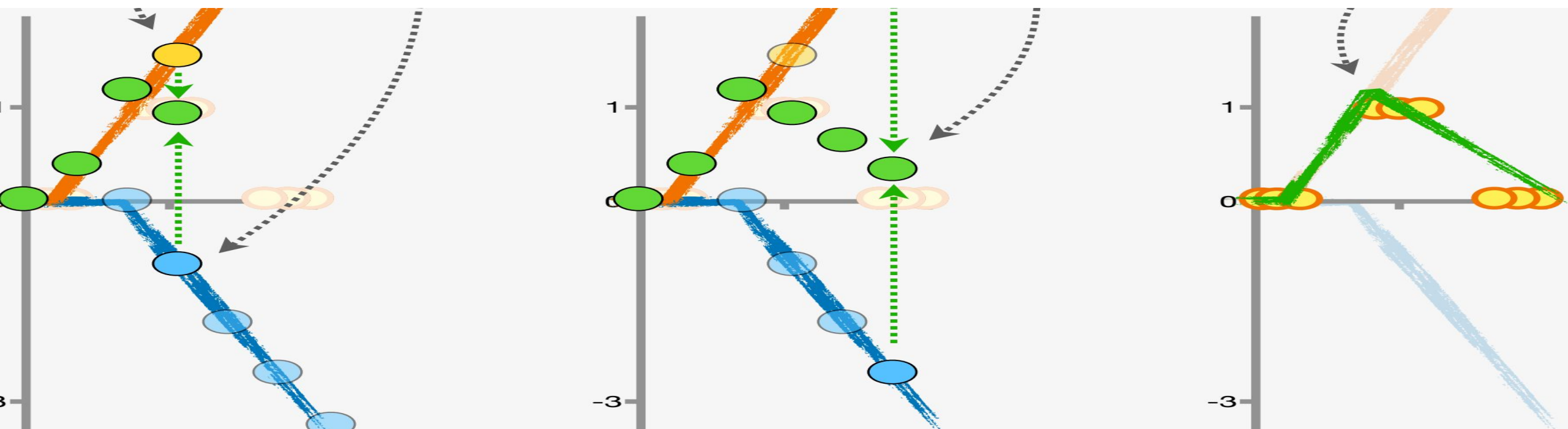
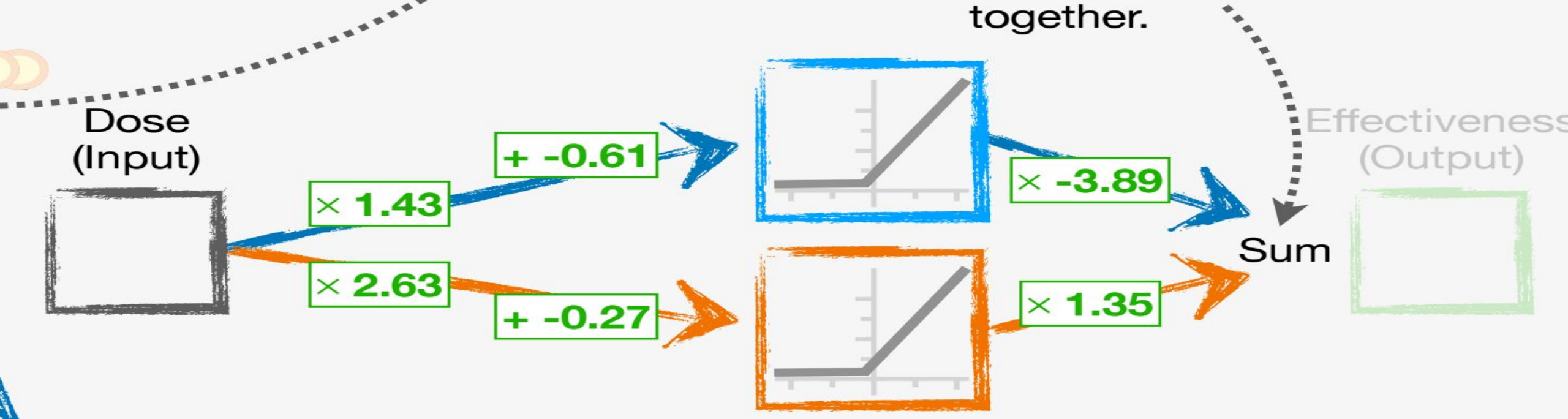


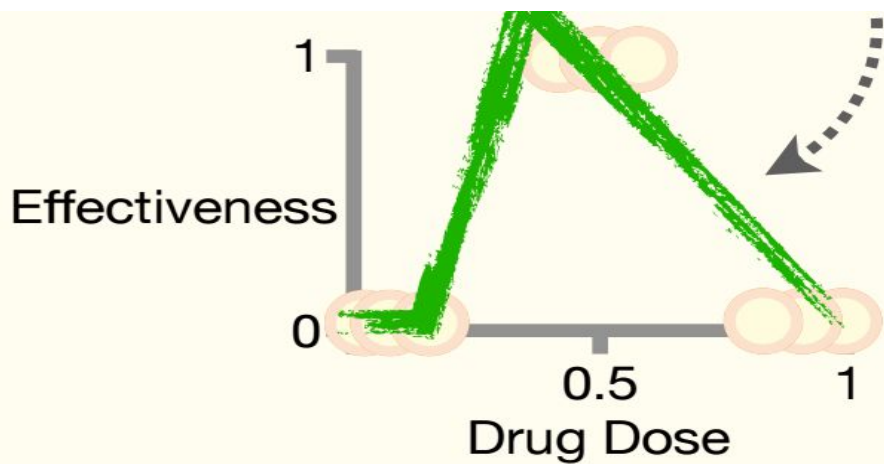


Drug Dose









...that this neural network uses to predict Effectiveness from Doses.

